Module 0: Magnetism Fundamentals

"The Missing Link" - Preparation for Electromagnetic Induction

1. The Nature of Magnets

Before understanding how to *generate* electricity from magnets (Induction), we must understand the magnetic field itself.

1.1 Properties of Permanent Magnets

- **Dipoles:** Every magnet has two poles: **North (N)** and **South (S)**. You can never isolate a single pole (no "monopoles").
- **Interaction:**
  - **Like** poles repel ($N \leftrightarrow N$, $S \leftrightarrow S$).
  - **Opposite** poles attract ($N \rightarrow\leftarrow S$).
- **The Field ($\vec{B}$):** An invisible vector field that exerts force.
  - **Direction:** OUT of North, INTO South.
  - **Strength:** Represented by the *density* of field lines. Closer lines = Stronger Field.
  - **Continuity:** Magnetic field lines form closed loops (they continue inside the magnet from S to N).

1.2 Common Shapes

Shape	Field Characteristics	Use Case
Bar Magnet	Non-uniform. Strongest at poles. Spreads out wide.	Compass needles, experiments.
U-Shape (Horseshoe)	**Uniform** field in the gap between poles.	Motors, generators (stator).

2. Electromagnetism: The Connection

Hans Christian Oersted (1820) discovered that **Electric Current creates Magnetic Fields.**

2.1 The Right Hand Grip Rule (RHR-1)

To find the direction of the magnetic field created by a current:

1. **Thumb:** Point in direction of Conventional Current (I).
2. **Fingers:** Curl around the wire.
3. **Result:** Your fingers point in the direction of the Magnetic Field ($\vec{B}$).

3. The "Big Three" Geometries

You will encounter these three specific shapes in almost every problem.

3.1 The Straight Wire

A long, straight wire carrying current I.

- **Shape of Field:** Concentric circles centered on the wire.
- **Formula:**

$$B = \frac{\mu_0 I}{2\pi r}$$

  - r: Distance from the wire (meters).

- *Note:* Field gets weaker as you move away ($1/r$).

3.2 The Flat Coil (Circular Loop)

A single wire loop (or flat stack of N loops) of radius R.

- **Shape of Field:** Doughnut-like. Field lines pass through the center.
- **Formula (at Center only):**

$$B_{center} = \frac{\mu_0 N I}{2R}$$

- *Note:* Notice the denominator is $2R$ (diameter), not $2\pi r$.

3.3 The Solenoid (The "Coil")

A long helix of wire. This is the most important shape for Induction.

- **Shape of Field:**
  - **Inside:** Straight, parallel lines. The field is **UNIFORM**.
  - **Outside:** Very weak (often considered zero in ideal problems), resembles a Bar Magnet.
- **Formula (Inside Core):**

$$B = \mu_0 \frac{N \cdot I}{L} = \mu_0 n I$$

  - N: Total turns.
  - L: Length of solenoid (m).
  - n: Turn density (N/L).
- **Uniformity:** Since B depends only on constants (μ_0, n, I), the field strength is the same everywhere inside the tube.

  Constant: $\mu_0 = 4\pi \times 10^{-7} T \cdot m/A$ (Permeability of Free Space)

4. Tech Connect: Solar & Automation 🛠️

Why does a technician care about $B = \mu_0 I / 2\pi r$?

1. DC Current Clamp Meters (Hall Effect) How do we measure 50A flowing from a solar string without cutting the wire? We clamp a meter around it.

- The meter measures the **Magnetic Field** (B) circling the wire.
- It uses the formula $I = \frac{B \cdot 2\pi r}{\mu_0}$ to calculate the current on the screen.

2. Relays & Contactors A relay is just a **Solenoid**.

- We send a small control current through the coil.
- It creates a magnetic field ($B = \mu_0 n I$).
- This field pulls a metal lever (armature) to close the high-power contacts.